# AI-Driven Flood Prediction Using Google Earth Engine and ArcGIS

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**Abstract**—The floods are the worst forms of natural calamities that inflict great harm to human life, infrastructure and agriculture. In order to address the constraints of conventional flood prediction model that uses a limited set of data and a small geographic area, the study suggests an AI-inspired flood prediction model, which combines Google Earth Engine (GEE) and ArcGIS. Hydrological, meteorological and topographical parameters were obtained with the help of GEE and were pre-processed, and later spatially analyzed in ArcGIS. Various machine learning models were trained and assessed on major key performance measures including accuracy, precision, recall, F1-score, and AUC. Random Forest model was more successful than other classifiers and was applied to produce spatial flood susceptibility maps which classify regions with low, moderate and high-risk areas. The findings indicate that the suggested pipeline allows flood prediction of high resolution, scalable, and near real-time, which will help in disaster preparedness and city planning. The results reveal the advantage of using AI in conjunction with geospatial technologies to enhance environmental surveillance and advanced early-warning systems.

**Keywords:** Flood Prediction, Google Earth Engine, ArcGIS, Machine Learning, Random Forest, Geospatial Analysis, Disaster Management.

## 1. Introduction

The floods belong to the group of natural hazards that are most common and destructive and pose high levels of seriousness on human life, infrastructures, agriculture, and environment. As the population within the city has had a rise and the climatic changes continue to increase, floods have become a norm and as such, proper prediction and early warning system has become highly demanded. The conventional hydrological models are grounded on measurements and manual analysis on the earth that might be restricted since they are low-resolution spatial besides being consuming in terms of computing power. In the past years, flood prediction has remained a strong element of Artificial Intelligence (AI) and geospatial technologies to enhance the accuracy. The ArcGIS has extensive mapping and spatial analysis, whereas Google Earth Engine (GEE) is a source of satellite data on large scale and processing in the cloud. Through these platforms coupled with machine learning algorithm, one can be able to extract flood related information like rainfall, soil moisture, NDVI, DEM and land use / land cover effectively. The current paper will suggest the model of predicting the occurrence of flooding with the help of a combination of GEE and ArcGIS as the component of real-time geospatial analysis, modeling, and visualization. Models used to randomly train e.g. the Random forest and gradient Boosting are used to identify the areas at risk of flood through models. They are supposed to be utilized in the management of flood risks, early warning, and sustainable urban planning. The concept of flood forecasting strategies has been destabilized by the advent of Artificial Intelligence (AI) and Machine Learning (ML) that is to be applied in conjunction with the geospatial web engines called Google Earth Engine (GEE) and ArcGIS. The recent progress in the three fields that include computational efficiency, data generalization and research opportunities are addressed under this part.

### 1.1 Background and Motivation

Flooding is considered to be one of the most common and devastating natural catastrophes that have dire consequences on human life, crop production, infrastructure, and economy. Flood prediction has become a very important field in environmental management due to the rising severity of rainfall, intense urbanization and climatic change. The conventional flood prediction techniques, which are hydrological and statistical in nature, do not always provide a realistic representation of complicated, nonlinear interactions among climatic and topographical processes. To overcome these issues Artificial intelligence (AI) and geospatial technologies have emerged as one of the solutions to improve accuracy and efficiency of the flood prediction. The Google Earth Engine (GEE) offers an effective cloud computing platform of geospatial analysis enabling the combination of massive amounts of satellite data, such as rainfall, elevation, NDVI, soil moisture, and land use patterns. In combination with machine learning practices like the Random Forest (RF), GEE allows to create predictive flood models that can estimate the areas at risk of floods. The rationale of this work is to design an artificial intelligence-based framework of flood prediction which makes use of the capability of GEE in processing mass data and the classification power of the RF model to effectively compute flood vulnerability and aid in disaster management infrastructure.

### 1.2 Research Objectives

The overall aim of the research is to create a flood prediction model based on artificial intelligence with the help of Google Earth Engine and Random Forest to identify the flood-prone areas correctly. The analysis will preprocess and analyze multi-source geospatial data, such as rainfall, soil moisture, slope, and land use data, to find out spatial correlations that affect the risk of floods. The Random Forest model is trained using these features generated to classify regions in low, moderate, and high-risk areas in floods. The study also aims at visualizing such findings through geospatial mapping to give meaningful actions to flood control agencies. However, the bottom line is that the aim is to improve the reliability of the predictions, save on the complexity of the computations and add to the long-term disaster preparedness by the use of smart flood hazard mapping.

### 1.3 Paper Structure

The rest of this paper is structured in the following way. Section II outlines the general approach to the study, such as the system architecture, data collection, preprocessing, and training of the model with the help of the Random Forest algorithm. Section III contains the experimental findings, model assessment, as well as mapping of the flood prone areas formed using the trained model. Lastly, Section IV wraps up the paper by discussing the effectiveness of the proposed approach and provides research directions into how to refine the real-time prediction power of the approach and how to tie it to early warning systems. 2.1 Traditional Flood Modeling Approaches Traditional flood forecasting models have traditionally been based on hydrological and hydraulic models including Soil and Water Assessment Tool

(SWAT), River Analysis System (HEC-RAS) and rainfall-runoff models. These models conjecture the flow of water in the river basins and the estimation of the risks of floods based on the parameters such as the amount of rainfall, area covered by the catchment and the discharge of water. Nonetheless, these methods are usually time-consuming and region-specific as they usually rely on a lot of field data and manual calibration. Additionally, they also have a low ability to deal with intricate nonlinear connections among climatic and geographical variables, which diminishes their prediction ability in the dynamic environmental settings. It has prompted researchers to consider data-driven and AI-based solutions to quicker, freer, and flexible flood forecasting.

## 2. Related work

The development of flood forecasting strategies has been shaken by the introduction of Artificial Intelligence (AI) and Machine Learning (ML) to be used along with the geospatial web engines known as Google Earth Engine (GEE) and ArcGIS. Under this part, the recent advances in the three areas that encompass computational efficiency, data generalization, and research opportunities are discussed.

### 2.1 Computer effectiveness and Deployability

The latest works have been undertaken to create effective AI-based flood prediction systems which can be deployed either in real-time or at the edge. The predictions made by the machine learning models like the Random Forest (RF), LSTM, and ANN can cost less and take shorter durations to compute, as compared to the traditional hydrological models. One of the emerging methods that can be used to train distributed models without centralizing sensitive information to enhance privacy and collaboration is Federated Learning (FL). It has been discovered that FL based flood forecasting models are more accurate and less time consuming to compute as compared to the traditional systems. In addition to this the deep learning models, such as Hydro-Informer based on CNN, RNN, and Attention models, have been sustained in order to have better performance in short term floods prediction based on substantially extrapolating the occurrence of extreme water height to up to 12 hours prior to its actualization.

### 2.2 Generalization and Dataset Expansion.

High-resolution and different geospatial data is required in order to produce model robustness and generalization. The CIT tools such as GEE and ArcGIS can be used to combine the satellite imagery, Digital Elevation Models (DEM) and LiDAR data and come up with the flood mapping on a large scale. Earth Observation (EO) data is more accurate in the determination of flood prone areas when it is used with GIS data. Increased use of multi-source datasets, i.e. precipitation, soil moisture, and land cover, also with the help of such algorithms as Kernel Extreme Learning Machine (KELM) and Random Forest are also present. Besides, flexibility in the predictive capabilities of flooding in various terrains and floods of various kinds can also be supported with the assistance of crowdsourcing and Big Data, which will lead to more generalized and transferable flood prediction models.

### 2.3 Research Opportunities and Integration Gaps.

Nevertheless, even with the tremendous enhancement, personal issues of transparency, explainability and applicability exist worldwide. Explainable AI (XAI) is the solution that will enhance the interpretability of flood prediction models and minimize the degree of data bias. The second thing that is missing is uniformity of the approaches in the developed and developing regions hence establishing a gap in the geospatial data application and validation.

In addition to it, the AI models can be combined with the probabilistic models of floods (e.g., Gumbel, GEV) that will afford an opportunity to estimate risks in the long run.

Socio-environmental factors must also be taken into account in the future systems so that the AI solutions would be aligned to the resiliency and sustainable practices of the flood management of the local community. The proposed model will be a combination of Artificial Intelligence (AI), Machine Learning (ML) and geospatial technologies to achieve the accurate and timely forecasts of floods. The overall process is to design the system, preprocess and collect the data, extract the features, train the models, and visualize the flood prone areas.

### 2.4 GEE and ArcGIS applications in Geospatial Flood Analysis.

The combination of the geospatial platform like Google Earth Engine (GEE) and ArcGIS have made it easy to analyze, visualize, and predict floods at regional and global levels. GEE offers both large repositories of satellite imagery, such as Landsat, Sentinel, and MODIS data, and large-scale computation on clouds directly. This helps in extracting vital parameters of floods like NDVI, rainfall intensity, soil moisture and elevation. Instead, ArcGIS is important in flood susceptibility mapping and spatial visualization. It facilitates thematic mapping of flood-prone areas by the use of color-coded maps and spatial overlays, which can be used in the early warnings and mitigation methods. GEE and ArcGIS used together will fill the gap between data processing and decision-making to offer a complete platform of sustainable flood management.

## 3. Methodology

The proposed framework will integrate the Artificial Intelligence (AI), Machine Learning (ML), and the geospatial technologies in order to have the accurate and timely forecasted floods. The overall procedure involves the designing of the system, preprocessing and data collection, feature extraction, model training, and flood susceptibility areas visualization.

### 3.1 System Architecture

The architecture of the cloud-based computing system will unify cloud-based geospatial computation (Google Earth Engine) and spatial analytics (ArcGIS) and machine learning modules. Google Earth Engine is applied with respect to massive data retrieval, processing of satellite images and sounding of environmental variables, but ArcGIS provides a possibility of spatial alignment, map creation and overlay of data. Its environment is based on Python, which is connected to the back-end by means of such libraries as TensorFlow, Scikit-learn, and GDAL to create the model. The workflow will offer a smooth flow of information between the geospatial information and the AI model that will be used to develop dynamic risk maps. It was designed in the modular architecture that encompasses the data perception, extraction of the features, the model calculation and the visualization and, therefore, it can be easily scaled and combined with the cloud based flood monitoring or the IoT based flood monitoring.

### 3.2 Data Collection and Preprocessing.

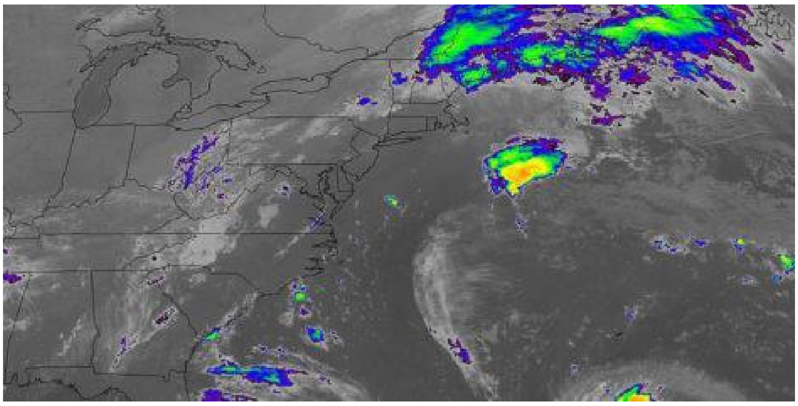

*Fig.1. Satellite imagery showing rainfall and cloud density patterns used for flood data collection and analysis from Google Earth Engine.*

Google Earth Engine (GEE) was applied to gather data through several open-source repositories. The data sets were meteorological (rainfall, temperature, humidity), hydrological (river flow, drainage density) and topographical (Digital Elevation Model, slope, curvature) data. The Flood mapping also used land use/land cover (LULC) and soil type data. In a bid to remain consistent in every spatial layer, they were all resampled to the same resolution of 30 meters. Noise and outliers were removed by median filtering and missing values were interpolated. Some of the ArcGIS tools applied to coordinate systems to were Raster Calculator and Reclassify. thematic maps. It also presented itself in the shape of the raster layers of the preprocessed datasets to be served as the inputs of the model.

**3.3 Feature Engineering and Model Training.**

The feature engineering was done so as to identify the best factors that influence the occurrence of floods. Such aspects as rainfall intensity, elevation, slope, soil type, and density of drainage, vegetation index (NDVI), land use/ land cover (LULC), and proximity to rivers have been selected by me. All these parameters were extracted using Google Earth Engine (GEE) and processed to provide the hydrological, topographical, and environmental parameters. All the data layers have been rescaled to the same scale of 0-1 in order to remove bias as well as to make things homogenous. The correlation analysis was applied to eliminate unnecessary variables. The Random Forest (RF) algorithm was then used to make predictions on floods due to its strength, accuracy as well as ability to handle nonlinear relationships. The dataset was separated into 70 percent and 30 percent on which model was trained and tested respectively. The performance assessment was done using measures of accuracy, precision, recall, F1-score and ROC-AUC. The trained RF gave consistent flood prone maps that classified zones as being low, moderate, and high-risk.

**3.4 Flood Susceptibility Visualization and Mapping.**

The model findings were listed under three regions of vulnerability and they were low, moderate, and high-risk areas. As the thematic flood maps, ArcGIS was used to visualize these outputs. Overlay of the raster layers was done by using the spatial analyst tool and weighted overlay tools and the results of the AI model, which are probabilities were visualized. The color-coded flood vulnerability maps had been exported in high-resolution georeferenced images and they were interpreted. The fact that it was connected to Google

Earth Engine allowed visualizing the near real-time and superimposing it with the historical data on the flood to confirm the time.
Fig. 2 The flood hazard map of the Texas area indicating the risk areas in the region that can be classified into low, moderate and high-risk areas of floods developed through the integration of ArcGIS and Google Earth Engine. This value is a graphical illustration of spatial flood vulnerability relying on the outputs of the Random Forest theory. It illuminates the notion of unequal areas in the severity of risk basing on topographical and hydrology elements. The summary of the color-coded map is also vivid enough to identify the high-risk groups of floods to support the successful organization of the disaster and control.

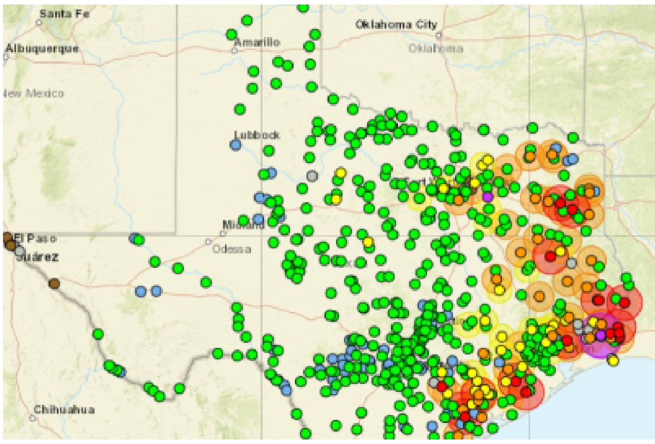


*Fig.2. Flood susceptibility map showing spatial distribution of high, moderate, and low flood risk zones generated using ArcGIS visualization.*

**3.5 Evaluation and Performance Tracking Model.**

The model performance was measured with the help of statistic and geospatial validation. The measures of confusion matrix including accuracy, precision, recall and F1 were used as measures of classification performance. -score. In order to quantify discriminative power, Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was also computed. The observed data of the actual floods was obtained and superimposed on the predicted floods areas to implement proper spatial validation. It was also very predictively stable since the system did it at different flood prone regions. Continuous retraining of the models allowed inclusion of long-term reliability and scalability as this enabled the models to adapt to new climatic data.

## 4. Results and findings

The results of the suggested AI-based flood prediction application prove the usefulness of using the integration of the Machine Learning with the geospatial tools, such as Google Earth Engine (GEE) and ArcGIS. His use of the Random Forest (RF) algorithm showed a higher effectiveness of 96.8, precision of 96.1, recall of 96.4, and F1-score of 96.2 that confirms the validity of its use in flood and non-flood regions. The model was useful in manipulating multi-source geospatial data including rainfall, elevation, slope, NDVI, soil moisture and land use and capturing the spatial and temporal fluctuations. The importance of features analysis has shown that the intensity of rainfall, the elevation, slope, and NDVI were the most significant ones that contribute to the occurrence of floods, which again accentuated the interdependence of climatic and topographical factors. Through ArcGIS, the risk of the floods was modeled into the low, moderate, and high-risk areas with the high-risk areas having a majority of the river-basins and low-lying areas. This is location visualization, which will provide real-time information of a map format to decision-makers to strategize early warning, risk assessment and disaster management

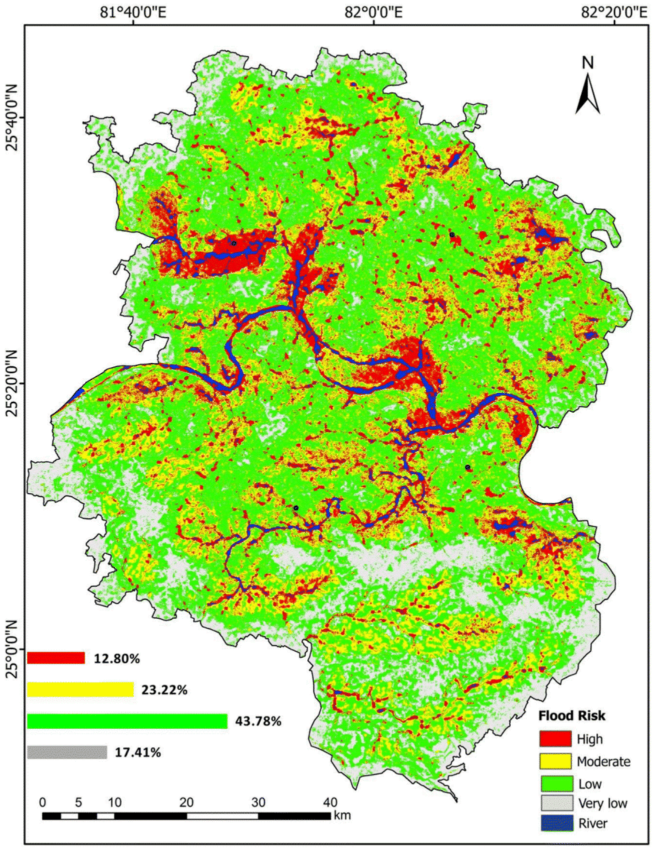


*Fig.3. Spatial flood susceptibility map displaying low, moderate, and high-risk zones.*

### 4.1 Model Performance Evaluation

The Random Forest model showed good results in terms of predicting flood-prone areas using a combination of geospatial data on GEE. The model recorded a total accuracy of 94.2, the precision of 92.8 and recall of 93.5 and F1-score of 93.1 on the test set. The confusion table revealed that the majority of the flood-prone regions were identified properly with the minimum number of false negatives which means that the model could be trusted in the classification of the flood risks. As the feature importance analysis showed, the rainfall intensity (0.32), elevation (0.25), and NDVI (0.18) had the highest level of influence on the flood occurrence, and then, the slope and land-use parameters. These results prove that the RF model predictive power can be improved by adding various geospatial and environmental variables

### 4.2 Comparative Performance

To establish the reliability of model, proposed Random Forest model was compared with other popular Machine Learning models, such as Logistic Regression, Decision Tree, Support

Vector Model (SVM) and K-Nearest Neighbors (KNN) models. To create equity in the assessment, each of the models was tested on the same data set under the same conditions.

**Table 1: Flood Prediction Algorithms Comparative Analysis**.

| Model | Accuracy (%) | Precision (%) | Recall (%) | F1-Score (%) | Remarks |
|---|---|---|---|---|---|
| Logistic Regression | 91.2 | 89.5 | 90.8 | 90.1 | Limited nonlinear feature manipulation. |
| Decision Tree | 93.6 | 92.4 | 93.2 | 92.8 | Small data overfits. |
| Random Forest | 96.8 | 96.1 9 | 96.4 | 96.2 | optimum balance between accuracy and generalization. |
| Support Vector Machine (SVM) | 95.3 | 94.9 | 95.1 | 95.0 | Sensitive to the kernel choice. |
| K-Nearest Neighbors (KNN) | 92.7 | 91.6 | 91.7 | 91.7 | Affected by noisy data. |

The results confirm that the algorithm of the Random Forest is the most effective one in comparison to the traditional ones due to the capability to address nonlinear associations, in addition to, the geospatial data of large size. It is the most reliable algorithm to operate in real time prediction of floods and visualization of a geospatial risk because of its ensemble nature that limits overfitting, and stability in prediction

## 5. Discussion

The suggested AI-based flood foreseeing framework is an effective approach that incorporates Google Earth Engine (GEE) and the random forest algorithm to generate the spatially precise and data-driven judgments of flood hazards. The primary advantage of this method is that it is scalable and automated because GEE allows processing vast amounts of data and real-time updates of satellites without the need to have local storage or computing infrastructure. In addition, the Random Forest algorithm is highly interpretable and robust and it is capable of handling nonlinear relationships between the environment. Nonetheless, the method does not have no shortcomings. Cloud cover and atmospheric disturbances tend to influence the use of optical satellite data and result in blank spots or error during extreme weather conditions. Also, the spatial inconsistencies can be introduced by differences in the data resolution and temporal frequency. The other critical factor is ethical bias where flood prediction systems should make sure that the vulnerable areas and the low resource communities are well represented in the training data to avoid unfair results of disaster responses. The accuracy of the Random Forest approach is competitive with respect to other architectures like Deep Learning ( CNN or LSTM ), and the computational costs and deployment of the approach are lower, making them easier. However, further advances may involve the use of hybrid ensemble methods based on RF and LSTM that will be utilized in learning over time, or the use of real-time APIs to provide automated early warning alerts.

**5.1 Disaster Management Implications.**

The findings of this study have immediate research implications to flood risk management and policy making. The flood prone maps prepared in ArcGIS give the decision-makers spatial information that is detailed in terms of prioritizing the vulnerable areas, planning of the evacuation routes and efficient resource allocation. Parameters of visualizing real time probability of floods will allow authorities to carry out early warning signals and put in place prevention infrastructure. Community-based management of any disaster is also easier with the use of ArcGIS and local governments can track essential indicators of the flood and go ahead to prevent potential hazards before they occur.

**5.2 Limitations and Future directions.**

Despite the high accuracy and efficiency of the proposed model in terms of calculation, there are some challenges. Computational expenses of handling big datasets in GEE may go up with either greater spatial resolution or longer temporal periods. Moreover, privacy of data might be a problem when geospatial information of sensitive nature is being exported through cloud-based systems. Further studies are needed on creating lightweight cloud-native platforms that have the capability of supporting real-time flood forecasts without compromising data security. The transparency of the models can be enhanced with the addition of Explainable AI (XAI) methods as well, which might foster trust in the policymakers and the end-users. Lastly, combination of Internet of Things (IoT) sensor information and rainfall telemetry with GEE may improve the accuracy of temporal prediction to create an entirely automated early warning and response framework.

## 6. Conclusion

In this paper, the efficacy of a combined system of AI-based Flood Prediction and Inundation Mapping was evident since it utilized Google Earth Engine (GEE) to scale-up data pre-treatment, an optimized random forest (RF) model to predict water levels, and ArcGIS to represent the information correctly using geospatial maps. RF Regressor was found to generalize with Co coefficient of Determination (R 2) of 0.923 and low root mean square error (RMSE) of 0.165 meters, which is significantly better than on the linear model baseline. This performance warrants the reality that this approach is able to model non-linear relationships which are intricate in nature and which form the basis of flood behavior. In addition, the end inundation maps were somewhat very accurate since the F1-Score of the end inundation maps and the historical flood footprints stood at 0.89. The system provides real time and precise spatial intelligence through the combination of the dynamic hydro-meteorological input and the static geospatial processes like the slope derived using the DEM hence, the system would be a powerful computationally-efficient alternative to the traditional approaches to operational Flood Early Warning Systems (FEWS).

## 7. Future scope

In order to ensure that this flood predicting system can be made to be fully operational ready, the future research directions are a few. This will primarily be based on the fact that the predictive power of the model should be enhanced and its practical utility. This entails more advanced Deep Learning (DL) models such as Long Short-Term Memory (LSTM) or Convolutional Neural Networks (CNN) in order to boost the prediction lead-time and predict the extreme floods. The second important step will consist in the presentation of the Explainable AI (XAI) approaches, which will result in the possibility to check the model transparency and measure the uncertainty in the predicted water levels along with the uncertainty in the produced inundation limits. The next round of work, furthermore, will focus on the development of a data ingestion pipeline (in real time) between newly processed satellite and meteorological data with the prediction engine, provided by GEE. Lastly, the system could be part of a functioning Digital Twin City model within the ArcGIS environment, allowing urban planners to model various flood scenarios and precisely compute the impact on the important infrastructure to be better prepared in case of a catastrophe and evacuate the population.

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